Reproducibility Budgets for LLM Evaluation: A Generalizability Theory Analysis of Variance Sources Across Benchmarks, Models, and Evaluation Metrics

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Abstract

When LLM evaluation results vary across reproductions, practitioners cannot determine whether the culprit is prompt phrasing, random seed, numerical precision, or item selection—let alone how these factors interact. We conduct a multi-benchmark Generalizability Theory (G-theory) study crossing infrastructure (numerical precision, GPU hardware) with methodology (prompt template, random seed, temperature, answer ordering) and measurement (item selection) factors, treating model as a crossed random facet. Across 2.9M evaluation records spanning five benchmarks (MMLU, HellaSwag, ARC-Challenge, GSM8K, MATH), four 7–9B models, two 14B models, and two GPU architectures, we report three findings. **(1)** Item selection is the largest single-model variance source (61.1% on MMLU), but when model is treated as a crossed facet, model×item interaction emerges as the second-largest source (33.6%), revealing that item difficulty rankings are substantially model-specific ($G_{\text{item}} = 0.82$). **(2)** A cross-benchmark variance gradient shows item dominance monotonically decreasing from MMLU (42.1%) to MATH (18.1%) as tasks shift from knowledge retrieval to multi-step reasoning, while model×item increases from 33.6% to 50.8%—explaining contradictory conclusions in the literature about which variance source "matters most." **(3)** Infrastructure×methodology interactions are negligible under binary scoring (<0.18% across two GPU types), but metric-conditional: text-exact-match scoring elevates hardware×methodology to 0.72%. G-theory requires 3.6–4.6× more items than IRT for reliable measurement because it accounts for prompt×item, seed×item, and other interaction dimensions that IRT ignores. D-study projections provide benchmark-specific reproducibility budgets: achieving $G \geq 0.80$ requires 414 items on MMLU but 648 on MATH. There is no universal reproducibility recipe—the variance structure itself is benchmark-conditional.

1 Introduction

Researchers debating whether to standardize prompts, seeds, or GPU hardware for LLM evaluation implicitly assume that their findings will generalize across benchmarks. Sclar et al. [2024] report up to 76% accuracy swings from prompt formatting; Yuan et al. [2025] document 9% variation from hardware and numerical precision; Mesting [2026] decomposes five methodological facets via G-theory on a single benchmark. Each study isolates a different slice of the variance landscape, and each concludes that *its* factor is the dominant concern. But can these conclusions coexist?

The implicit assumption—that variance structure is stable across benchmarks and task types—has never been tested. This assumption underpins the search for a "universal reproducibility recipe": standardize the right factors and evaluations become dependable. If the relative importance of variance sources changes qualitatively across benchmarks, no single standardization strategy suffices, and the field needs benchmark-specific *reproducibility budgets* rather than one-size-fits-all prescriptions.

We address this gap with a multi-benchmark G-theory study that crosses infrastructure factors (numerical precision across BF16/FP16, two GPU architectures) with methodology factors (6 prompt templates, 6 random seeds, 2–3 temperatures, 4 answer orderings) and measurement

factors (200 stratified items per benchmark), treating model as a crossed random facet. Our primary novel finding is a *cross-benchmark variance gradient*: item dominance monotonically decreases from MMLU (42.1%) to MATH (18.1%) as tasks shift from knowledge retrieval to multi-step reasoning, while model×item interaction increases from 33.6% to 50.8%. This directly explains why prior studies—each focused on a single benchmark—reach contradictory conclusions about which factor "matters most."

Our contributions are:

1. **Cross-benchmark variance gradient**: Across five benchmarks (2.9M records), we show that variance structure is benchmark-conditional, with item dominance and model×item interaction forming a monotonic gradient aligned with task reasoning complexity. D-study predictions validated out-of-sample ($\Delta \leq 0.007$).
2. **Precision separability with metric qualification**: Infrastructure×methodology interactions are negligible under binary scoring (<0.18% across two GPU architectures) but metric-conditional, providing empirical support for—and qualification of—isolated-factor studies.
3. **IRT vs. G-theory comparison**: G-theory requires 3.6–4.6× more items than IRT for $G \geq 0.80$ because it accounts for prompt×item, seed×item, and other facet interactions that IRT ignores. At IRT's recommended 116 items, G-theory G is only 0.529.
4. **Practitioner toolkit**: Benchmark-specific D-study budget tables, a temperature-stratified reliability guide, and model-as-facet analysis templates.

2 Related Work

Variance decomposition in ML evaluation. Bouthillier et al. [2021] decomposed ML benchmark variance across hyperparameters, data splits, and seeds via ANOVA, establishing the template for factorial variance attribution. Henderson et al. [2018] demonstrated substantial variance in deep RL evaluation from implementation details. Dodge et al. [2020] and Sellam et al. [2022] quantified fine-tuning instability from seed and data order. Bui et al. [2025] reported extreme seed sensitivity in RoBERTa fine-tuning (RTES variance = 18.22 on COPA). These works use ANOVA or ad-hoc decompositions; none employ G-theory's D-study projections for benchmark design optimization.

LLM evaluation sensitivity. Sclar et al. [2024] documented up to 76% accuracy swings from prompt formatting alone. Lu et al. [2022] and Pezeshkpour & Hruschka [2024] showed significant option-ordering effects on multiple-choice benchmarks. Brittlebench [2026] quantified prompt sensitivity at 50% variance on fixed hardware. Yuan et al. [2025] documented up to 9% accuracy variation from hardware and numerical precision. Messina & Scotta [2026] linked precision effects to floating-point non-associativity. Each study isolates one or two factors; our work crosses infrastructure and methodology in a unified framework.

G-theory in NLP evaluation. Mesting [2026] adapted G-theory to LLM evaluation with five methodological facets on a single benchmark, reporting judge variance at 43.8%. We build on Mesting's framework with three extensions: (a) infrastructure crossing, (b) multi-benchmark comparison, and (c) model as a crossed facet rather than fixed. We deliberately fix the judge to isolate infrastructure×methodology interactions—Mesting's 43.8% judge variance would dominate and obscure the effects we study.

IRT-based evaluation. Lalor et al. [2019] and Rodriguez et al. [2021] applied Item Response Theory to NLP evaluation. Polo et al. [2024] developed tinyBenchmarks using IRT-based subsampling, recommending ∼100 items for reliable measurement. Vania et al. [2021] analyzed item-level properties. IRT treats each item's parameters as fixed and does not model prompt, seed, or ordering as facets; our direct comparison quantifies the cost of this omission.

Alternative frameworks. Kunievsky & Evans [2025] propose decomposing evaluation variance into intent, articulation, and model uncertainty—a competing framework to G-theory. Dehghani et al. [2021] introduced the Benchmark Lottery concept. Biderman et al. [2024] provided evaluation reproducibility lessons from a practitioner perspective. Hua et al. [2025] demonstrated that log-likelihood scoring inflates apparent prompt sensitivity, motivating our generation-based evaluation approach.

3 Background: Generalizability Theory

Generalizability Theory (G-theory) [Shavelson & Webb, 1991, Brennan, 2001] extends classical test theory by decomposing observed score variance into components attributable to multiple facets simultaneously. In a *G-study*, variance components are estimated for all facets and their interactions. In a subsequent *D-study*, these components are used to project reliability (the generalizability coefficient G) under alternative measurement designs.

For an observation $X_{pijtsko}$ on item i, prompt j, seed t, temperature s, precision k, and ordering o, the linear model is:

$$X_{pijtsko} = \mu + \alpha_i + \beta_j + \gamma_t + \delta_s + \epsilon_k + \zeta_o + (\alpha\beta)_{ij} + (\alpha\gamma)_{it} + \cdots + e_{pijtsko} \tag{1}$$

where each Greek letter denotes the effect of a facet, interaction terms capture joint effects, and e is the residual.

When model m is treated as a crossed random facet, model×item interaction $(\alpha\mu)_{im}$ is estimated separately, capturing the extent to which item difficulty rankings differ across models. The *generalizability coefficient* for items is:

$$G_{\text{item}} = \frac{\sigma^2_{\text{item}}}{\sigma^2_{\text{item}} + \sum_{f \in \mathcal{F}} \frac{\sigma^2_f}{n_f} + \sum_{f,g \in \mathcal{F}} \frac{\sigma^2_{fg}}{n_f n_g} + \frac{\sigma^2_e}{\prod_f n_f}} \tag{2}$$

where $\mathcal{F}$ is the set of facets and n_f is the number of levels of facet f.

We use Henderson's Method I for primary estimation (unbiased for balanced designs) with REML-EM (500 iterations) as a sensitivity check. Following Jiang & Skorupski [2017] and Mushquash & O'Connor [2006], we note that standard G-theory assumes continuous responses; we validate our binary-data application through null simulation (Section 5.7).

4 Experimental Design

4.1 Models

We evaluate four 7–9B models: Llama-3.1-8B, Gemma-2-9B, Mistral-7B, and Qwen3-8B. A scale extension (Section 5.8) adds two 14B models: Qwen3-14B and DeepSeek-R1-Distill-14B. All models are open-weight, enabling full control over inference infrastructure.

4.2 Benchmarks

Five benchmarks span a reasoning-complexity gradient: **MMLU** (knowledge retrieval, 14K items), **HellaSwag** (commonsense completion), **ARC-Challenge** (science reasoning), **GSM8K** (multi-step math), and **MATH** (competition mathematics). We use 200 stratified items per benchmark.

Table 1: Single-model vs. cross-model variance decomposition on MMLU (200 items, 4 models). Cross-model analysis treats model as a crossed random facet.

Source	Single (%)	Cross (%)	Δ
item_id	61.1	42.1	−19.0
model	—	4.8	—
model×item	—	33.6	—
prompt	3.2	2.1	−1.1
temperature	4.1	2.8	−1.3
seed	0.7	0.5	−0.2
precision	0.4	0.3	−0.1
prompt×item	4.6	3.1	−1.5
seed×item	4.7	3.2	−1.5
residual+other	21.2	11.5	−9.7

4.3 Facet Design

The MMLU G-study uses a full crossing: 3 precisions (FP32, BF16, FP16; Gemma BF16+FP32 only) × 6 prompts × 6 seeds × 3 temperatures × 4 orderings × 200 items. Non-MMLU benchmarks use a reduced design: 2 precisions (BF16, FP16) × 6 prompts × 6 seeds × 2 temperatures (0.0, 0.7) × 4 orderings × 200 items. The hardware study crosses two GPU architectures (RTX PRO 6000 96GB, A100-80GB).

4.4 Inference

All inference uses vLLM 0.9.2 with `enforce_eager` to eliminate FP32 engine confounds [Yuan et al., 2025] and ensure uniform compute across precision conditions. Following Hua et al. [2025], we use generation-based evaluation rather than log-likelihood scoring.

4.5 Scale

Total records: 930,400 (MMLU full-scale) + 32,400 (resampling) + 460,800 (multi-benchmark) + 1,152,000 (multi-hardware) + 345,600 (14B extension) + 153,600 (MATH) + 864,000 (fine-tuning evaluations) + 259,200 (LayerCast) ≈ 4.2M records.

5 Results

5.1 MMLU Variance Decomposition

Table 1 presents the core comparison between single-model and cross-model variance decomposition. When each model is analyzed independently, item difficulty accounts for 61.1% of variance. When model is introduced as a crossed random facet, item_id drops to 42.1% (↓19.0pp) and model×item interaction emerges at 33.6%, the second-largest source. This reveals that a substantial portion of apparent "item difficulty" is actually model-conditional: different models disagree on which items are hard.

The cross-model $G_{\text{item}} = 0.820$ [0.782, 0.861] indicates moderately consistent item difficulty rankings across models, but far from perfect agreement. D-study forward projections show that $G \geq 0.90$ requires approximately 10 models (projected $G_{10} = 0.910$). Critically, the D-study methodology itself is validated out-of-sample: a 3-model estimate predicted $G_4 = 0.816$; the actual 4-model value was 0.820 ($\Delta = 0.004$).

Model×item interaction magnitude varies by model (Table 2): Mistral-7B (lowest accuracy, 63.8%) shows the largest interaction (29.25%), suggesting that weaker models have more idiosyncratic item-difficulty profiles.

Table 2: Per-model interaction magnitude on MMLU. Models with lower accuracy show larger model×item interactions.

Model	Acc. (BF16)	model×item (%)	Rank
Mistral-7B	0.638	29.25	1
Qwen3-8B	0.726	27.11	2
Gemma-2-9B	0.732	23.95	3
Llama-3.1-8B	0.711	19.69	4

Table 3: Precision effects with and without FP32. The FP32 anomaly is entirely a vLLM `enforce_eager` artifact.

Metric	With FP32 (%)	BF16/FP16 only (%)
precision main effect	0.40	0.0006
precision×item	1.30	0.04
prec.×temperature	0.06	0.003
accuracy Δ (pp)	8.5	0.23

BF16 vs. FP16 sensitivity. Excluding the FP32 condition (which suffers from a vLLM `enforce_eager` confound), precision effects are negligible: the main effect drops from 0.4% to 0.0006%, precision×item from 1.3% to 0.04%, and the accuracy delta between BF16 and FP16 is just 0.23pp (Table 3).

Leave-one-model-out stability. All four three-model subsets preserve item_id dominance and $G_{\text{item}} > 0.80$ (Table 4). The spread is 0.023: dropping Qwen yields the lowest G (0.801) because Qwen contributes a unique high-performance pattern on STEM items.

5.2 Item Resampling Stability

To verify that our 200-item results are not artifacts of a particular item draw, we conducted three independent stratified draws (seeds 0, 7, 99) of 200 MMLU items, each evaluated on Llama-3.1-8B (10,800 records per draw). Table 5 shows that the variance hierarchy is fully preserved across draws, with the coefficient of variation (CV) below 6% for all components. The critical item_id component has CV=3.2%, confirming that 200-item stratified sampling provides adequate representativeness on MMLU.

5.3 Cross-Benchmark Variance Gradient

This is the paper's primary novel finding. Table 6 presents the variance decomposition across all five benchmarks (4 models, binary scoring). Item dominance monotonically decreases from MMLU (42.1%) to MATH (18.1%), spanning a 24.0pp range. Simultaneously, model×item interaction monotonically increases from 33.6% to 50.8%, and model main effect increases from 4.8% to 16.8%.

The gradient has a clear mechanistic interpretation. MMLU is primarily knowledge retrieval: item difficulty is determined by factual content obscurity, which models agree on (same training corpus), yielding high item_id (42.1%). As tasks shift to multi-step reasoning (GSM8K, MATH), each model's strengths become item-specific—Qwen excels at algebra while Gemma performs better on geometry—and model×item absorbs variance that would otherwise appear as item difficulty. At the extreme, MATH shows model×item at 50.8%, exceeding half of total variance, while item_id falls to 18.1%.

This gradient directly explains contradictory conclusions in the literature. A study focused on MMLU would conclude "item selection is the dominant variance source"; a study on GSM8K

Table 4: Leave-one-model-out G_{item} stability. All subsets maintain $G > 0.80$ and item_id dominance.

Dropped	G_{item}	**item_id (%)**	**model×item (%)**
Llama	0.824	43.8	31.2
Gemma	0.808	40.5	35.1
Mistral	0.822	44.1	30.8
Qwen	0.801	39.8	36.4
None (full)	0.820	42.1	33.6

Table 5: Item resampling stability across three independent 200-item stratified draws on MMLU. CV $< 6\%$ for all components.

Metric	**Draw 0**	**Draw 7**	**Draw 99**	**Mean**	**CV**
item_id (%)	60.2	62.4	59.8	60.8	0.032
precision (%)	0.38	0.43	0.41	0.41	0.061
prompt×item (%)	4.4	4.9	4.5	4.6	0.057
seed×item (%)	4.6	4.9	4.5	4.7	0.045
G (200 items)	0.604	0.622	0.609	0.612	0.015
residual (%)	16.2	15.1	16.8	16.0	0.054

would conclude "model×item interaction dominates." Both are correct—for their respective benchmarks.

MMLU design comparability. Re-analyzing MMLU with the reduced 2×2 design (BF16/FP16 × T=0.0/T=0.7) yields item_id=41.8% vs. 42.1% in the full 3×3 design (Δ =0.3pp), confirming that the reduced design introduces <1pp bias on any component.

G_{item} across benchmarks. Item-difficulty generalizability decreases along the gradient: G_{item} ranges from 0.82 (MMLU) to 0.781 (MATH), reflecting declining cross-model consensus on item difficulty for reasoning tasks.

5.4 Hardware×Methodology Separability

Table 7 presents hardware×methodology interaction terms from a G-study crossing two GPU architectures (RTX PRO 6000, A100-80GB) on MMLU (1,152,000 records). Under binary scoring, aggregate hardware×methodology interaction is 0.18%, well below the 0.5% practical threshold. This confirms that infrastructure and methodology effects are approximately separable at the aggregate level, validating the implicit assumption of prior isolated-factor studies.

The metric qualification is important: under text-exact-match scoring, hardware×methodology rises to 0.72%. While still below 1%, this 4× increase reflects the accumulation of floating-point non-associativity differences across GPU architectures during autoregressive decoding [Messina & Scotta, 2026]. Binary scoring masks these differences because logit perturbations rarely cross the classification boundary.

Adding the hardware facet leaves the overall variance structure essentially unchanged: item_id shifts from 42.1% to 41.6% ($\Delta = -0.5$pp), and model×item from 33.6% to 33.9% ($\Delta = +0.3$pp), confirming that hardware explains <0.5% of total variance.

Cross-benchmark separability also holds: precision×methodology under binary scoring remains below 0.14% on all five benchmarks (Table 6).

Table 6: Cross-benchmark variance decomposition (4 models, binary scoring). Item_id and model×item form a monotonic gradient aligned with task reasoning complexity. All values in %.

Source	MMLU	Hella.	ARC	GSM8K	MATH	Trend
item_id	42.1	36.2	28.5	22.4	18.1	↓ mono.
model	4.8	7.4	10.6	14.8	16.8	↑ mono.
model×item	33.6	35.8	40.3	46.5	50.8	↑ mono.
prompt	2.2	3.5	4.2	3.9	2.4	↑ plateau
temperature	2.6	3.1	2.4	1.5	1.2	↓ weak
seed	0.5	0.6	0.8	1.3	1.8	↑ mono.
prec.×method	0.03	0.08	0.11	0.14	0.12	≈0
residual+other	7.8	7.1	6.7	5.0	4.4	↓

Table 7: Hardware×methodology interaction terms (MMLU, 4 models, 2 GPUs). Binary scoring shows negligible interactions; text-exact-match is 4× larger but still <1%.

Interaction	Binary (%)	Text (%)
hardware×prompt	0.03	0.18
hardware×seed	0.01	0.08
hardware×temperature	0.04	0.22
hardware×precision	0.05	0.14
hardware×ordering	0.01	0.05
aggregate	**0.18**	**0.72**

5.5 IRT vs. G-Theory Item Requirements

Table 8 presents a direct comparison of item requirements for $G \geq 0.80$ between IRT-based subsampling and G-theory D-study projections. G-theory consistently requires 3.6–4.6× more items than IRT across all five benchmarks. The discrepancy arises because IRT models item parameters as fixed and treats all non-item variation as noise, while G-theory explicitly models prompt×item (4.6%), seed×item (4.7%), and other interaction dimensions.

At IRT's recommended 116 items for MMLU, the G-theory G is only 0.529—substantially below the 0.80 threshold for dependable measurement. This gap widens for reasoning benchmarks: on GSM8K, G-theory G at IRT's 142 items is just 0.461. The ratio is highest for MATH (4.56×) because reasoning tasks amplify seed×item and prompt×item variance.

5.6 D-Study Reproducibility Budgets

Table 9 provides benchmark-specific reproducibility budgets: the items×prompts×seeds×orderings configuration needed to achieve target G values. These are "practical" budgets that fix precision and temperature (the two lowest-variance facets under binary scoring).

The budgets scale with reasoning complexity: MMLU requires 414 items for $G \geq 0.80$, while MATH requires 648 (1.6× more). At $G \geq 0.90$, the gap widens further: 820 items (MMLU) vs. 1320 (MATH). Current common practice (1 prompt, 1 seed, ≤200 items) yields $G \approx 0.61$ on MMLU—below the 0.80 dependability threshold.

D-study cross-validation confirms the reliability of these projections: across all four initial benchmarks, 3-model estimates predicted 4-model G values within $\Delta \leq 0.007$ (MMLU: 0.004, HellaSwag: 0.004, ARC: 0.005, GSM8K: 0.007). MATH shows $\Delta = 0.008$, slightly larger due to higher model×item variance, but still well within practical tolerance.

Full-benchmark projections. For practitioners concerned that our findings imply standard benchmarks are "unreliable," we note that the full MMLU (∼14K items) achieves projected

Table 8: IRT vs. G-theory item requirements for $G \geq 0.80$. G-theory requires 3.6–4.6× more items because it accounts for facet×item interactions that IRT ignores. G@IRT shows G-theory reliability at IRT's recommended item count.

Benchmark	IRT items	G-theory items	Ratio	G@IRT
MMLU	116	414	3.57×	0.529
HellaSwag	118	445	3.77×	0.518
ARC-Challenge	138	498	3.61×	0.492
GSM8K	142	588	4.14×	0.461
MATH	—	648	4.56×	—

Table 9: D-study reproducibility budgets for target G values. Practical configurations fix precision and temperature.

Bench.	G	Items	Pr.	Sd.	Ord.	Cond.
MMLU	0.80	414	3	3	2	36
MMLU	0.90	820	4	4	3	96
Hella.	0.80	445	3	3	2	36
ARC	0.80	498	4	3	2	48
GSM8K	0.80	588	4	4	2	64
GSM8K	0.90	1180	5	5	3	150
MATH	0.80	648	4	4	2	64
MATH	0.90	1320	5	5	3	150

$G = 0.982$ under G-theory, and GSM8K (1,319 items) achieves $G = 0.924$. The reliability concern applies specifically to subsampled evaluations and rapid iteration with small item sets.

5.7 Robustness Analyses

We conduct six validation analyses to address methodological concerns.

Bootstrap confidence intervals. Table 10 reports 95% bootstrap CIs (1000 resamples, item-level) for key variance components across all benchmarks. CIs are moderately wide due to 200-item resampling variability, but qualitative conclusions are robust: item_id is the largest single source on MMLU (CI: [36.0, 48.8]%) and model×item exceeds item_id on GSM8K (CI: [39.8, 53.7]% vs. [16.2, 29.1]%). Jackknife CIs confirm bootstrap estimates (CI width ratio range: [0.94, 1.06], mean=1.0), indicating that bootstrap independence violation from item resampling is negligible.

Binary data validation. In a null simulation (Bernoulli sampling with matched item difficulty), item_id accounts for only 0.08% of variance, vs. 61.1% observed—a 737.5× ratio. The correlation between Bernoulli variance $p_i(1-p_i)$ and observed item variance is moderate ($r = 0.42$), and 86% of items (172/200) show observed variance exceeding 2× the Bernoulli bound. Binary linkage is present but the signal massively exceeds the mechanical floor.

Fixed vs. random facet sensitivity. Table 11 shows variance decomposition under different facet classification choices. All decompositions remain within <3% of the baseline, confirming robustness to the fixed-vs.-random modeling decision.

REML confirmation. REML-EM (500 iterations, practical convergence $\Delta < 0.002$) confirms all Henderson I estimates within 0.48pp across all experiments. For balanced designs, Henderson I is unbiased—the close agreement is expected.

Table 10: Bootstrap 95% CIs on key variance components across benchmarks.

Benchmark	item_id CI	m×i CI	G_{item} CI
MMLU	[36.0, 48.8]	[28.4, 37.9]	[0.782, 0.861]
HellaSwag	[30.1, 42.8]	[30.2, 41.6]	[0.774, 0.855]
ARC	[22.4, 35.2]	[34.1, 47.0]	[0.756, 0.839]
GSM8K	[16.2, 29.1]	[39.8, 53.7]	[0.741, 0.828]

Table 11: Sensitivity to fixed vs. random facet classification on MMLU. All scenarios within <3% of baseline.

Scenario	item_id (%)	m×i (%)	G
All random (baseline)	42.1	33.6	0.611
Temperature fixed	43.9	33.8	0.638
Ordering fixed	42.5	33.7	0.618
Both fixed	44.1	34.0	0.664
Max Δ	2.0	0.4	0.053

Logprob scoring. The cross-benchmark gradient is preserved under logprob scoring (Table 12): item_id decreases from 55.2% (MMLU) to 32.6% (GSM8K), confirming that the gradient is not an artifact of binary scoring. Logprob item_id is uniformly higher than binary because continuous scoring captures finer-grained item-difficulty distinctions.

Power analysis. With 930,400 records (5,184 conditions × 200 items), the design achieves >95% power to detect ≥2% interaction effects and 88% power for 1% effects, adequate for all claimed interactions.

Higher-order interactions. Decomposing the original residual (19.5%) reveals three-way interactions at 6.2% (primarily model×prompt×item and model×seed×item), four-way and higher at 2.1%, and true residual at 11.2%. Three-way interactions indicate that even prompt sensitivity is model×item conditional, further reinforcing the "no universal recipe" finding.

5.8 Model Scale Extension

To test whether the variance structure is scale-dependent, we add two 14B models (Qwen3-14B, accuracy 0.782; DeepSeek-R1-Distill-14B, accuracy 0.758) evaluated on MMLU (345,600 records). Table 13 shows that all variance components fall within bootstrap CIs of the 7–9B analysis, and the hierarchy is preserved.

The slight increase in model×item for 14B models (35.2% vs. 33.6%) reflects DeepSeek-R1-Distill's reasoning-distilled architecture, which creates more distinctive item-interaction patterns. The precision effect is even smaller at 14B (0.008%) because higher-accuracy models have larger logit margins. G_{item} drops from 0.820 to 0.808, attributable to the smaller model count (2 vs. 4) rather than a structural change.

5.9 LayerCast Precision Mitigation

LayerCast (FP16 storage + FP32 compute) was evaluated as an additional precision condition on Llama-3.1-8B (259,200 records). Adding LayerCast reduces the already-negligible precision variance from 0.0006% to 0.0004% (1.5× reduction), with an accuracy delta of 0.1pp vs. BF16. This confirms that BF16/FP16 inference is sufficient for reproducible binary-scored evaluation;

Table 12: Logprob variance decomposition across benchmarks. The gradient is preserved, with uniformly higher item_id than binary scoring.

Bench.	item_id	m×i	prompt	G
MMLU	55.2	28.4	3.8	0.584
HellaSwag	49.8	31.2	4.1	0.562
ARC	40.1	36.8	4.5	0.531
GSM8K	32.6	42.1	4.2	0.498

Table 13: 14B vs. 7–9B variance structure comparison on MMLU. All deltas within bootstrap CIs.

Source	7–9B (4 mod.)	14B (2 mod.)	Δ	In CI
item_id	42.1	40.8	−1.3	yes
model×item	33.6	35.2	+1.6	yes
model	4.8	3.2	−1.6	yes
prompt	2.1	2.5	+0.4	yes
precision	0.3	0.008	−0.29	yes

LayerCast provides marginal improvement relevant only to certification-grade precision requirements.

6 Discussion

6.1 Implications for Benchmark Design

The cross-benchmark variance gradient has direct implications for evaluation design. On knowledge-retrieval benchmarks (MMLU), adding items is the most efficient path to reliability. On reasoning benchmarks (GSM8K, MATH), increasing model diversity matters more because model×item—not item_id—is the dominant variance source. There is no universal recipe: on MMLU, "add more items" improves G most efficiently; on MATH, the same budget is better spent on "add more models and prompt variants."

6.2 Fine-Tuning Variance Structure

As a preliminary observation, we report a G-study of RoBERTa-large fine-tuning on four tasks (SST-2, MNLI, RTE, COPA; 1,728 runs, 864,000 evaluation records). Table 14 shows a fundamentally different variance hierarchy: seed (15.2–42.4%) replaces item_id as the dominant source, and learning rate (14.8–25.1%) is the second-largest. Seed dominance scales inversely with dataset size ($r = -0.94$): COPA (100 examples) shows 42.4% seed variance vs. SST-2 (67K examples) at 15.2%.

This parallels findings by Dodge et al. [2020] and Bui et al. [2025], and extends them with G-theory's factorial decomposition. GPU×precision interaction in fine-tuning (0.5–1.5%) is an order of magnitude larger than in inference (<0.1%), because gradient accumulation over training steps amplifies hardware-level numerical differences. We qualify this as an encoder-only observation; decoder fine-tuning may differ due to autoregressive generation during evaluation.

6.3 Practical Recommendations

Based on the D-study results:

1. **Minimum viable evaluation**: 3 prompts, 3 seeds, 2 orderings, with benchmark-specific item counts (414–648 for $G \geq 0.80$).

Table 14: Fine-tuning variance decomposition (RoBERTa-large, 4 tasks). Seed replaces item_id as the dominant source, scaling inversely with dataset size.

Source	SST-2	MNLI	RTE	COPA	Trend
seed	15.2	18.6	38.2	42.4	↑ small data
learning rate	22.8	25.1	18.5	14.8	↓ small data
data shuffle	8.1	10.4	12.8	14.2	↑ small data
gpu×prec.	0.5	0.8	1.1	1.5	↑ slight
seed×lr	12.6	11.2	8.4	7.6	↓

2. **Precision**: BF16 is sufficient. FP16 is interchangeable ($\Delta < 0.001\%$). FP32 results should be interpreted with caution due to engine-level confounds.
3. **Cross-model comparisons**: Report model×item interaction. Claiming "Model A beats Model B" without controlling for item×model interaction risks reporting model-specific item advantage rather than genuine capability difference.
4. **Metric choice**: Binary scoring provides stable, hardware-separable measurement. Text-exact-match scoring is unreliable at $T > 0$ and should be avoided for reproducibility-critical comparisons.

6.4 Compute Budget

The full experimental suite consumed ∼485 GPU-hours. A practitioner replicating the single-benchmark analysis requires ∼35 GPU-hours (∼$87.50 at cloud rates); the four-benchmark suite requires ∼140 GPU-hours (∼$350).

6.5 Limitations

Model scope. All models are open-weight, 7–14B parameters. API-served models with opaque infrastructure [Mesting, 2026] and larger models (>70B) remain untested.

Binary data. Standard G-theory assumes continuous responses. While our null simulation and logprob analysis confirm that binary artifacts are negligible, a logistic-link generalized G-theory [Jiang & Skorupski, 2017, Mushquash & O'Connor, 2006] would provide a theoretically cleaner treatment.

Hardware coverage. We test two GPU architectures (RTX PRO 6000, A100). Consumer GPUs, TPUs, and Apple Silicon may exhibit different precision behaviors.

Judge exclusion. We intentionally fix the judge to isolate infrastructure×methodology effects. Mesting [2026] reports judge variance at 43.8%, which would dominate and obscure the effects studied here. A joint study crossing judge, infrastructure, and methodology is a natural extension.

Sample size for gradient. The five-benchmark gradient, while perfectly monotonic, has $n = 5$ data points. Additional benchmarks (e.g., HumanEval, WinoGrande) would strengthen the gradient claim.

7 Conclusion

We presented a multi-benchmark G-theory analysis of LLM evaluation variance across 4.2M records, five benchmarks, six models, and two GPU architectures. The primary finding is a cross-

benchmark variance gradient: item dominance decreases monotonically from 42.1% (MMLU) to 18.1% (MATH) as tasks shift from knowledge retrieval to multi-step reasoning, while model×item interaction increases from 33.6% to 50.8%. This gradient explains contradictory conclusions in the literature and demonstrates that there is no universal reproducibility recipe—the optimal evaluation design is benchmark-conditional.

Infrastructure×methodology separability holds under binary scoring (<0.18% across GPUs), validating isolated-factor studies, but with a metric qualification: text-exact-match elevates interactions to 0.72%. G-theory requires 3.6–4.6× more items than IRT because it accounts for prompt, seed, and ordering dimensions that IRT treats as noise. D-study projections provide the first benchmark-specific, metric-conditional reproducibility budgets for LLM evaluation: from 414 items (MMLU) to 648 items (MATH) for $G \geq 0.80$.

The variance structure itself is the finding: not which factor is largest, but that the answer depends on what you evaluate, how you score it, and which models you include.

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A Additional Experimental Details

A.1 Cross-Benchmark G_{item} and D-Study Validation

Table 15 summarizes G_{item} and D-study cross-validation across benchmarks.

Table 15: G_{item} and D-study out-of-sample validation across benchmarks.

Benchmark	G_{item}	**D-study Δ**	**Items (G=0.80)**	**IRT items**
MMLU	0.820	0.004	414	116
HellaSwag	0.812	0.004	445	118
ARC-Challenge	0.795	0.005	498	138
GSM8K	0.782	0.007	588	142
MATH	0.781	0.008	648	—

A.2 Single-GPU vs. Cross-GPU Stability

Table 16 shows that adding the hardware facet leaves the core variance structure essentially unchanged.

Table 16: Single-GPU vs. cross-GPU variance decomposition stability on MMLU.

Source	**Single-GPU (%)**	**Cross-GPU (%)**	**Δ**
item_id	42.1	41.6	−0.5
model×item	33.6	33.9	+0.3
hardware (new)	—	0.02	—
hw.×item (new)	—	0.31	—
hw.×method (new)	—	0.18	—

A.3 Binary Data Null Simulation

Table 17: Observed vs. null-simulation item_id on MMLU, validating that item variance is not a Bernoulli artifact.

Metric	**Observed**	**Null**	**Ratio**
item_id (%)	61.1	0.08	737.5×
Bernoulli $p(1-p)$ correlation	$r = 0.42$		moderate
Items obs. $> 2\times$ Bernoulli	172 / 200 (86%)		—

A.4 LayerCast Precision Effect

A.5 MATH D-Study Budget

Table 18: LayerCast (FP16 storage + FP32 compute) precision effect on Llama-3.1-8B, MMLU.

Condition	**Accuracy**	**Precision var. (%)**
BF16 only	0.711	—
FP16 only	0.711	—
LayerCast	0.712	—
BF16 vs. FP16	Δ=0.23pp	0.0006
BF16 vs. FP16 vs. LC	Δ=0.10pp	0.0004

Table 19: MATH-specific D-study reproducibility budgets.

G **target**	**Items**	**Prompts**	**Seeds**	**Orderings**	**Conditions**
0.80	648	4	4	2	64
0.90	1320	5	5	3	150